

# Soyeb Pervez Jim

Software Engineer

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## Professional Summary

Software engineer with remote experience building production LLM agents, semantic retrieval, and multi-tenant SaaS for e-commerce. Co-author on two Elsevier journal papers applying physics-informed deep learning to industrial systems; author of C++/Qt desktop software with 313 GitHub stars. English at CEFR C1.

## Technical Skills

**Languages:** TypeScript, Python, C++, QML, JavaScript, Swift, Lua, Bash

**AI/ML:** LLM agents, tool calling, retrieval-augmented generation (RAG), vector search, embeddings, AI SDK, OpenAI/Anthropic/Google models, PyTorch, ONNX, scikit-learn, CNN/TCN, biosignal processing

**Backend:** Node.js, Hono, Express, Next.js, React, Drizzle, Sequelize, Postgres, Vitess, Redis, Stripe, Qt 6/QML

**Infrastructure:** Docker, GitHub Actions, Traefik, Dokploy, Hetzner, Cloudflare R2, Vercel, k6, Playwright, Nix

## Professional Experience

### Software Engineer

*Fuego.io*

July 2025 – Present

*Remote*

- Led development of the tool-calling chat assistant behind a multi-tenant Shopify commerce platform: streaming responses, product search, recommendations, and reviews returned as structured tool results.
- Built semantic product search over live catalogs, with a scheduled re-embedding job keeping every tenant's index current.
- Automated per-brand context generation, so a new tenant gets a customer-care knowledge base from its own site with no manual authoring.
- Built the admin surface and metering: conversation analytics, token-usage tracking, and Stripe billing across tenants.

## Selected Projects

### HyprFM

*C++, Qt 6, QML, Wayland*

- Keyboard-first file manager with Miller columns, split panes, async transfers, device mounting, and live-reloading TOML themes; QML presentation split from C++ backend services, shipped on AUR, Flatpak, AppImage, and Nix. 313 stars, 9 outside contributors, roughly 350 unique cloners a fortnight, and an independent [LinuxLinks review](#).

### Neural Network X-Ray

*Next.js, React, TypeScript, ONNX Runtime*

- Browser visualizer showing activations across all 13 CNN layers as users draw Latin or Bengali characters, inference running locally through WebAssembly. Trained the 146-class model on roughly 980,000 EMNIST and BanglaLekha images, published on [Hugging Face](#).

## Research & Publications

**Physics-Informed Multi-Task Neural Network for Leak Detection in Multiphase Pipelines** *Published, 2026*  
*Process Safety and Environmental Protection* (Elsevier) | [doi.org/10.1016/j.psep.2026.109609](https://doi.org/10.1016/j.psep.2026.109609)

- PCE-PINN, a multi-task network predicting leak presence, diameter, and location from inlet/outlet pressure and flow alone:  $F1 = 1.00$  detection,  $R^2 = 0.906$  and  $0.941$ , a 16.5% location gain over the best classical baseline.

**Physics-Informed Embed-to-Control with Per-Producer MLP Heads for CO<sub>2</sub> Storage Surrogate Modelling**  
*Fuel Processing Technology* (Elsevier), submitted 2026 *Under review*

- Replaced the shared linear well model in an E2CO reservoir surrogate with per-producer MLP heads, trained under PINN residuals on the simulator's own finite-volume operator: saturation RMSE  $-38\%$ , pressure RMSE  $-35\%$ , per-well  $R^2$  from  $0.82$  to  $0.91$  in 16 min on one T4.

**Challenge papers.** AI4Pain 2026, ACII Grand Challenge (under review): two-stage physiological pain classifier, 0.572 test accuracy against a 0.398 baseline. MUMIDC, ABC2026: CNN2D-TCN over EEG, EMG, and IMU, 87.9% macro F1 under 16-fold mixed LOSO.

## Education & Achievements

**BSc in Electrical and Electronic Engineering**, University of Dhaka

2021 – 2026 (*expected*)

Final-year thesis on abnormal activity recognition | EF SET English certificate: C1

**Competitive programming.** [Codeforces](#) peak 1714 (Expert), 769 problems solved over 121 rated rounds. ICPC Dhaka Regional finalist twice: 37th/308 (2024), 37th/225 (2023). [National Collegiate Programming Contest](#) 27th/196; Meta Hacker Cup round 2 in 2023 and 2024.

**Olympiad.** Bangladesh Mathematical Olympiad national first runner-up (2017) and second runner-up (2016); co-authored [Ganit Olympiader Hatekhori](#), an olympiad primer (Swapno '71, 2024). Published [worked olympiad solutions](#) to 6.5K followers.